

Math 220A Practice Midterm

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Problem 1

Show that all finite p -groups are solvable.

Solution: Given G a p -group, with $|G| = p^n$ for some $n \in \mathbb{N}$, we'll proceed by induction on the power n .

First, for $n = 1$, since $|G| = p$ implies that G is cyclic, hence abelian, which is trivially solvable.

Now, suppose for $k < n$, all p -group with order p^k is solvable. To show that for $|G| = p^n$ implies G is solvable, there are several observations:

1. G has Nontrivial center:

If consider the conjugation action of G on itself, then based on the class equation, one has the following:

$$|G| = |Z(G)| + \sum_{i \in I} |\text{Conj}(g_i)| \quad (1.1)$$

where $Z(G)$ denotes the center of G (which every of them has singleton conjugation class, due to the fact that the elements are central), and each $i \in I$ corresponds to a g_i with distinct nontrivial conjugation class $\text{Conj}(g_i)$. Then, since $|G| = p^n$, the left side is divisible by p ; also, on the right side, since $|\text{Conj}(g_i)| < |G|$, and by orbit stabilizer theorem one has each $|\text{Conj}(g_i)|$ divides $|G| = p^n$. Then, with $|\text{Conj}(g_i)| \neq 1$ (due to the fact that $\text{Conj}(g_i)$ is non-singleton), one must have $|\text{Conj}(g_i)| = p^l$ for some positive integer $l < n$, so the sum $\sum_{i \in I} |\text{Conj}(g_i)|$ is divisible by p also (since each $|\text{Conj}(g_i)|$ is divisible by p). As a result, $|Z(G)| = |G| - \sum_{i \in I} |\text{Conj}(g_i)|$ is divisible by p , hence $Z(G) \neq \{e\}$.

As a result, one has $|Z(G)| = k$ for some positive integer $k < n$, which the quotient group $G/Z(G)$ has order $|G|/|Z(G)| = p^{n-k}$ with $0 < n - k < n$, hence $G/Z(G)$ is a p -group with power $n - k < n$, which is solvable. Hence, there exists the following chain of abelian tower:

$$\{1\} = K'_0 < K'_1 < \dots < K'_s = G/Z(G), \quad K'_i/K'_{i-1} \text{ is abelian} \quad (1.2)$$

2. Abelian Tower pulls back to an abelian tower:

Given projection $\pi : G \twoheadrightarrow G/Z(G)$, suppose $N' \leq M' \leq G/Z(G)$ are subgroups satisfying $N' \trianglelefteq M'$, then let $N := \pi^{-1}(N')$ and $M := \pi^{-1}(M')$, it's clear that $N \leq M$, with $\pi(N) = N'$ and $\pi(M) = M'$, due to surjectivity of π .

Now, we claim that $N \trianglelefteq M$, and $M/N \cong M'/N'$: If consider the projection restricting to M , then $\pi : M \twoheadrightarrow M'$; now, if consider a further projection $\varphi : M' \twoheadrightarrow M'/N'$, then the composition $\varphi \circ \pi : M \twoheadrightarrow M'/N'$ has kernel characterized by $\ker(\varphi \circ \pi) = N$ (since $n \in \ker(\varphi \circ \pi)$ iff $\varphi(\pi(n)) = 0$ iff $\pi(n) \in N'$ iff $n \in \pi^{-1}(N') = N$), hence this shows that $N \trianglelefteq M$, moreover by First Isomorphism Theorem, since $\varphi \circ \pi$ is surjective, one also has $M'/N' \cong M/N$.

As a result, since $\{1\} = K'_0 < \dots < K'_s = G/Z(G)$ is an abelian tower, for each index i , let $K_i := \pi^{-1}(K'_i)$, then one has the following tower (Note that $\pi^{-1}(\{1\}) = Z(G)$):

$$Z(G) = K_0 < \dots < K_s = G \quad (1.3)$$

And, for each index i , one has $K_i/K_{i-1} \cong K'_i/K'_{i-1}$ based on statement in 2. So, with each K'_i/K'_{i-1} being abelian, $Z(G) = K_0 < \dots < K_s = G$ forms a chain of subgroups, with each consecutive quotient being abelian.

Finally, since $Z(G)$ is abelian, concatenating $\{1\}$ the trivial group at the end completes an abelian tower, hence showing that G is solvable. And, this completes the induction.

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Problem 2

Find all conjugacy classes of S_3 , S_4 , and D_8 .

Solution: First, for S_3 and S_4 , there is a small lemma to use:

Lemma

Given S_n , two elements are in the same conjugacy class, iff they have the same cycle type (or, if their disjoint cycle decomposition have the same collection of lengths).

Note: For instance, in S_5 both $(1)(234)(5)$ and $(2)(4)(135)$ both has cycle type 1, 1, 3. Which, the number of cycle types for S_n is determined by the ways of partitioning n into positive integers.

Proof:

$\Rightarrow$: First, suppose $\sigma, \gamma \in S_n$ are conjugates of each other, say $\sigma = \tau\gamma\tau^{-1}$ for some $\tau \in S_n$. Then, let $\gamma = (a_1^1 \dots a_{k_1}^1) \dots (a_1^l \dots a_{k_l}^l)$ be the disjoint cycle decomposition of γ (where the a_j^i runs through all elements in $\{1, \dots, n\}$ to include all cycles in γ). Then, notice that the conjugation of each cycle in γ based on τ has the following property:

$$\forall i \in \{1, \dots, n\}, \quad \tau \circ (a_1^j \dots a_{k_j}^j) \tau^{-1} \cdot i = \begin{cases} i & i \notin \{\tau^{-1} \cdot a_1^j, \dots, \tau^{-1} \cdot a_{k_j}^j\} \\ \tau(a_{r+1}^j) & i = \tau^{-1}(a_r^j) \end{cases} \quad (2.1)$$

Where, if $r = k_j$, then $a_{r+1}^j := a_1^j$. So this concludes that $\tau \circ (a_1^j \dots a_{k_j}^j) \tau^{-1} = (\tau(a_1^j) \dots \tau(a_{k_j}^j))$, as a k_j -cycle.

As a result, we have the following:

$$\begin{aligned}\sigma &= \tau\gamma\tau^{-1} = (\tau(a_1^1 \dots a_{k_1}^1)\tau^{-1}) \dots (\tau(a_1^l \dots a_{k_l}^l)\tau^{-1}) \\ &= (\tau(a_1^1) \dots \tau(a_{k_1}^1)) \dots (\tau(a_1^l) \dots \tau(a_{k_l}^l))\end{aligned}\quad (2.2)$$

Since τ is a bijection on $\{1, \dots, n\}$, then it sends each disjoint cycles to another disjoint cycles with the same length, so σ has the same cycle type as γ .

$\Leftarrow$: Suppose $\sigma, \gamma \in S_n$ have the same cycle type, say under disjoint cycle decomposition, we have the following:

$$\gamma = (a_1^1 \dots a_{k_1}^1) \dots (a_1^l \dots a_{k_l}^l), \quad \sigma = (b_1^1 \dots b_{k_1}^1) \dots (b_1^l \dots b_{k_l}^l) \quad (2.3)$$

Then, each given that $i \neq i'$ or $j \neq j'$, one has $a_j^i \neq a_{j'}^{i'}$ and $b_j^i \neq b_{j'}^{i'}$, and each a_j^i, b_j^i runs through all elements in $\{1, \dots, n\}$.

Now, define a map $\tau : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ by $\tau(a_j^i) = b_j^i$. Then, τ forms a bijection (since each b_j^i corresponds to a unique preimage, namely a_j^i). And, for each cycle in γ above, it satisfies the following:

$$\tau(a_1^i \dots a_{k_i}^i)\tau^{-1} = (\tau(a_1^i) \dots \tau(a_{k_i}^i)) = (b_1^i \dots b_{k_i}^i) \quad (2.4)$$

Hence, under conjugation of τ , one has the following:

$$\begin{aligned}\tau\gamma\tau^{-1} &= (\tau(a_1^1 \dots a_{k_1}^1)\tau^{-1}) \dots (\tau(a_1^l \dots a_{k_l}^l)\tau^{-1}) \\ &= (\tau(a_1^1) \dots \tau(a_{k_1}^1)) \dots (\tau(a_1^l) \dots \tau(a_{k_l}^l)) \\ &= (b_1^1 \dots b_{k_1}^1) \dots (b_1^l \dots b_{k_l}^l) = \sigma\end{aligned}\quad (2.5)$$

Hence, γ, σ are in the same conjugacy class. $\square$

As a result, if one can find all methods of partition n into positive integers, it uniquely determines all cycle types.

1. For S_3 , since $3 = 1 + 1 + 1 = 1 + 2 = 3$ (the only three ways of partition 3), then it corresponds to conjugacy class $C_{1,1,1} = \{e\}$ (since it's the only permutation with three 1-cycles), $C_{1,2} = \{(12), (13), (23)\}$ (since they all have one 2-cycle, and another 1-cycle / fixing 1 element), and $C_3 = \{(123), (132)\}$ (all the 3-cycles).
2. For S_4 , since $4 = 1 + 1 + 1 + 1 = 1 + 1 + 2 = 2 + 2 = 1 + 3 = 4$ (the only five ways of partition 4), then it corresponds to conjugacy class $C_{1,1,1,1} = \{e\}$ (the only permutation with four 1-cycles), $C_{1,1,2} = \{(12), (13), (14), (23), (24), (34)\}$ (all permutation with one 2-cycles, and fixing two elements / two 1-cycles), $C_{2,2} = \{(12)(34), (13)(24), (14)(23)\}$ (all permutations with two disjoint 2-cycles), $C_{1,3} = \{(123), (132), (124), (142), (134), (143), (234), (243)\}$ (all 3-cycles, while fixing 1 elements), and $C_4 = \{(1234), (1243), (1324), (1342), (1423), (1432)\}$ (all 4-cycles).
3. For dihedral group D_8 , since it's generated by rotation σ and a reflection τ , that satisfy $\sigma^4 = \tau^2 = e$, and $\tau\sigma = \sigma^3\tau$. Then, $D_8 = \{e, \sigma, \sigma^2, \sigma^3, \tau, \tau\sigma, \tau\sigma^2, \tau\sigma^3\}$. Also, there are the following:
 - First, since $\sigma \cdot \sigma^2 = \sigma^2 \cdot \sigma$ by definition, and $\tau\sigma^2 = \sigma^3\tau\sigma = \sigma^3 \cdot \sigma^3\tau = \sigma^2\tau$, then σ^2 commutes with all generators, hence is central. So, besides $C_1 = \{e\}$, $C_2 = \{\sigma^2\}$ is another singleton conjugacy class.

- Since $\tau\sigma = \sigma^3\tau$, then $\tau\sigma\tau^{-1} = \sigma^3$, so $\{\sigma, \sigma^3\}$ are contained in the same conjugacy class. Also, $\sigma \cdot \sigma \cdot \sigma^{-1} = \sigma$, this shows that $C_3 = \{\sigma, \sigma^3\}$ forms a conjugacy class (since under finite conjugation of τ and σ , it still stays in C_3).
- Since $\tau(\tau\sigma^3)\tau^{-1} = \sigma^3\tau = \tau\sigma$, so $\{\tau\sigma, \tau\sigma^3\}$ are contained in the same conjugacy class. Also, $\sigma(\tau\sigma^3)\sigma^{-1} = \sigma(\tau\sigma^3)\sigma^3 = \sigma\tau\sigma^2 = \sigma \cdot \sigma^2\tau = \sigma^3\tau = \tau\sigma$, showing $C_4 = \{\tau\sigma, \tau\sigma^3\}$ does form a conjugacy class (since under finite conjugation of τ and σ , it still stays in C_4 , so it's invariant under D_8 conjugation action).
- Since $\sigma(\tau\sigma^2)\sigma^{-1} = \sigma\tau\sigma = \sigma \cdot \sigma^3\tau = \tau$, then $\{\tau, \tau\sigma^2\}$ are contained in the same conjugacy class. Also, $\tau(\tau\sigma^2)\tau^{-1} = \sigma^2\tau = \tau\sigma^2$, showing $C_5 = \{\tau, \tau\sigma^2\}$ does form a conjugacy class (since under finite conjugation of τ and σ , it still stays in C_5 , so it's stable under conjugation of D_8).

Which, one can count that the above classes for each groups does contain all possible elements, hence must be all the conjugacy classes.

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Problem 3

Let R be a ring and S be an R -algebra. Show that the functor from the category of R -algebras to itself given by $\otimes_R S$ is left adjoint to the $\text{Hom}_R(S, _)$. Conclude that tensor product preserves surjections and the Hom functor preserves injections. Given an example when tensor fails to preserve injectivity.

Solution:

Note: Over $\mathbf{R-Mod}$, this is true; but over $\mathbf{R-Alg}$, because $\text{Hom}_R(S, L)$ doesn't even have a well-defined addition structure, this category fails to be self-enriched.

The reason why $\text{Hom}_R(S, L)$ doesn't have a module structure in general, is because given $s, s' \in S$ and $f, g \in \text{Hom}_R(S, L)$, one has the following:

$$(f + g)(s \cdot s') = f(s \cdot s') + g(s \cdot s') = f(s) \cdot f(s') + g(s) \cdot g(s') \quad (3.1)$$

But, if $f + g$ is also an R -algebra homomorphism, one must satisfy the following:

$$\begin{aligned} (f + g)(s \cdot s') &= (f + g)(s) \cdot (f + g)(s') = (f(s) + g(s)) \cdot (f(s') + g(s')) \\ &= f(s) \cdot f(s') + f(s) \cdot g(s') + g(s) \cdot f(s') + g(s) \cdot g(s') \end{aligned} \quad (3.2)$$

So, the two are equal iff $f(s) \cdot g(s') + g(s) \cdot f(s') = 0$, which is not true in general.

Here instead of working over $\mathbf{R-Alg}$, we'll provide a proof of $S \in \mathbf{R-Mod}$ case.

First, given any $T, L \in \mathbf{R-Mod}$, let $\text{Bil}(T \times S, L)$ denote all R -bilinear maps $B : T \times S \rightarrow L$. Then, if consider the bilinear projection $\pi : T \times S \rightarrow T \otimes_R S$, it provides a natural bijection between $\text{Bil}(T \times S, L) \xrightarrow{\sim} \text{Hom}_R(T \otimes_R S, L)$, by $B \mapsto \bar{B}$, where it satisfies $\bar{B} \circ \pi = B$.

The universality of tensor product guarantees the existence and uniqueness of $\bar{B}$, hence the map is injective; moreover, any $\bar{B}' \in \text{Hom}_R(T \otimes_R S, L)$ has $\bar{B}' \circ \pi \in \text{Bil}(T \times S, L)$ due to the bilinearity of π , hence the map is also surjective. Finally, this map is a R -linear map also, since given any $a, b \in R$ and $B, B' \in \text{Bil}(T \times S, L)$, if $\bar{B}, \bar{B}' \in \text{Hom}_R(T \otimes_R S, L)$ satisfies $\bar{B} \circ \pi = B, \bar{B}' \circ \pi = B'$, then $(a \cdot \bar{B} + b \cdot \bar{B}') \circ \pi = a \cdot B + b \cdot B'$, showing $(a \cdot \bar{B} + b \cdot \bar{B}') = a \cdot \bar{B} + b \cdot \bar{B}'$.

Now, we can also establish a bijection between $\text{Bil}(T \times S, L)$ and $\text{Hom}_R(T, \text{Hom}_R(S, L))$, by mapping $B \mapsto F_B$, where for all $t \in T$, one has $F_B(t) = B(t, _) \in \text{Hom}_R(S, L)$ (since B is bilinear, fixing an entry ensures it's a linear map). This map is injective, since if $B, B' \in \text{Bil}(T \times S, L)$ satisfies $F_B = F_{B'}$, then for all $(t, s) \in T \times S$, one has $B'(t, s) = F_{B'}(t)(s) = F_B(t)(s) = B(t, s)$, so $B = B'$; it's surjective, since given any R -linear map $F : T \rightarrow \text{Hom}_R(S, L)$, one can define a bilinear map $B_F : T \times S \rightarrow L$ by $B_F(t, s) := F(t)(s)$ (where $F(t) \in \text{Hom}_R(S, L)$). so, this establishes a bijection $\text{Bil}(T \times S, L) \cong \text{Hom}_R(T, \text{Hom}_R(S, L))$ (which is also R -linear, since any $a, b \in R$ and $B, B' \in \text{Bil}(T \times S, L)$ has $F_{aB+bB'}(t) = (aB + bB')(t, _) = a \cdot B(t, _) + b \cdot B'(t, _) = a \cdot F_B(t) + b \cdot F_{B'}(t)$).

So, these two establishes the natural R -linear isomorphism $\text{Hom}_R(T \otimes_R S, L) \cong \text{Bil}(T \times S, L) \cong \text{Hom}_R(T, \text{Hom}_R(S, L))$, showing $_ \otimes_R S$ is left adjoint to $\text{Hom}_R(S, _)$.

As a result, since left adjoint functors preserves colimit (for instance, cokernel in $\mathbf{R}\text{-Mod}$), while right adjoint preserves limit (for instance, kernel in $\mathbf{R}\text{-Mod}$), hence given a surjective map $f : M \twoheadrightarrow N$, one has $f \otimes \text{id}_S : M \otimes_R S \twoheadrightarrow N \otimes_R S$ also being surjective (since cokernel of f is $N \rightarrow 0$, so with tensor preserving cokernel, the cokernel of $f \otimes \text{id}_S$ is $N \otimes_R S \rightarrow 0 \otimes_R S = 0$, hence $f \otimes \text{id}_S$ is surjective).

Similarly, given an injective map $g : P \hookrightarrow L$, one has $g \circ _ : \text{Hom}_R(S, P) \hookrightarrow \text{Hom}_R(S, L)$ being injective (since the kernel of g is $0 \rightarrow P$, then with $\text{Hom}_R(S, _)$ preserving kernel, the kernel of $g \circ _$ is $0 = \text{Hom}_R(S, 0) \rightarrow \text{Hom}_R(S, P)$, showing $g \circ _$ is injective).

Finally, the tensor functor is not left exact (or doesn't preserve injections) in general: Consider the inclusion map $\iota : \mathbb{Z} \hookrightarrow \mathbb{Q}$ (view both as abelian groups, or $\mathbb{Z}$ -modules), if tensor with $\mathbb{Z}/2\mathbb{Z}$, one has $\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$ (since as $\mathbb{Z}$ -modules, tensor with $\mathbb{Z}$ doesn't change the structure); however $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} = 0$, since given any $q \in \mathbb{Q}$ and $n \in \mathbb{Z}/2\mathbb{Z}$, one has $q \otimes n = (2 \cdot \frac{q}{2}) \otimes n = \frac{q}{2} \otimes (2 \cdot n) = \frac{q}{2} \otimes 0 = 0$ (since $2 \cdot n = 0$ in $\mathbb{Z}/2\mathbb{Z}$). So, the induced map $\iota \otimes \text{id}_2 : \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} = 0$ is no longer injective (since it's a zero map from a nontrivial $\mathbb{Z}$ -module).

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Problem 4

Let V and W be two (finite dimensional) representations of a finite group G over k . Show that the representations $\text{Hom}_k(V, W)$ is isomorphic to the tensor representation $V^\vee \otimes_k W$.

Solution: First, we'll show that they're isomorphic as vector spaces: Recall that $\dim k = 1$ (as k -vector space), $\dim V^\vee = \dim V$ (under finite dimensional case), and $\dim \text{Hom}_k(V, U) = (\dim V) \cdot (\dim U)$ for any finite dimensional k -vector space V, U . Then, using the natural isomorphism $\text{Hom}_k(V^\vee \otimes_k W, k) \cong \text{Hom}_k(V^\vee, \text{Hom}_k(W, k))$ (based on the adjunction of $\otimes_k W$ and $\text{Hom}_k(W, _)$ functors), we have the following:

$$\begin{aligned}
 \dim V^\vee \otimes_k W &= \dim \text{Hom}_k(V^\vee \otimes_k W, k) \\
 &= \dim \text{Hom}_k(V^\vee, \text{Hom}_k(W, k)) \\
 &= (\dim V^\vee) \cdot (\dim \text{Hom}_k(W, k)) \\
 &= (\dim V) \cdot ((\dim W) \cdot (\dim k)) \\
 &= (\dim V) \cdot (\dim W) \\
 &= \dim \text{Hom}_k(V, W)
 \end{aligned} \tag{4.1}$$

Since the dimension matches up, the two are isomorphic k -vector spaces.

Now, consider the map $B : V^\vee \times W \rightarrow \text{Hom}_k(V, W)$, by $B(\varphi, w) = T_{\varphi, w}$, where for all $v \in V$, one has $T_{\varphi, w} = \varphi(v) \cdot w \in W$ (since $\varphi \in V^\vee$, so $\varphi(v) \in k$). Which, the map B is well-defined, since $T_{\varphi, w} : V \rightarrow W$ is linear (given any $a, b \in k$ and $v, u \in V$, we have $T_{\varphi, w}(av + bu) = \varphi(av + bu)w = a \cdot (\varphi(v)w) + b \cdot (\varphi(u)w) = a \cdot T_{\varphi, w}(v) + b \cdot T_{\varphi, w}(u)$).

Also, notice that B is bilinear: given any $a, b \in k, \varphi, \phi \in V^\vee$, and $w, z \in W$, we have the following (for all $v \in V$):

$$\begin{aligned} B(a\varphi + b\phi, w) &= T, \quad T(v) = (a\varphi + b\phi)(v) \cdot w = (a \cdot \varphi(v) + b \cdot \phi(v)) \cdot w \\ &= a \cdot (\varphi(v)w) + b \cdot (\phi(v)w) = a \cdot T_{\varphi, w}(v) + b \cdot T_{\phi, w}(v) \\ &= a \cdot B(\varphi, w)(v) + b \cdot B(\phi, w)(v) \end{aligned} \quad (4.2)$$

$$\begin{aligned} B(\varphi, aw + bz) &= S, \quad S(v) = \varphi(v)(aw + bz) = a \cdot (\varphi(v)w) + b \cdot (\varphi(v)z) \\ &= a \cdot T_{\varphi, w}(v) + b \cdot T_{\varphi, z}(v) = a \cdot B(\varphi, w)(v) + b \cdot B(\varphi, z)(v) \end{aligned} \quad (4.3)$$

Which, $B(a\varphi + b\phi, w) = a \cdot B(\varphi, w) + b \cdot B(\phi, w)$, and $B(\varphi, aw + bz) = a \cdot B(\varphi, w) + b \cdot B(\varphi, z)$, showing B is bilinear.

Then, using the universality of tensor product, this uniquely factors to a linear map $\bar{B} : V^\vee \otimes_k W \rightarrow \text{Hom}_k(V, W)$.

Finally, we claim that $\bar{B}$ is an isomorphism, and also a morphism of G -representations, which implies that $\bar{B}$ is an isomorphism as morphism of G -representations.

1. Since the domain and codomain of $\bar{B}$ have the same (finite) dimension, to show it's an isomorphism of vector spaces, it suffices to show surjectivity.

Fix a basis $v_1, \dots, v_n \in V$, and $w_1, \dots, w_m \in W$ (say $n = \dim V$, $m = \dim W$). Define the linear map $D_{ij} : V \rightarrow W$ by $D_{ij}(v_l) = \delta_{jl} w_i$ (where δ_{jl} is the Kronecker delta). Then, for all $T \in \text{Hom}_k(V, W)$, given $T(v_l) = \sum_{i=1}^m a_{il} w_i$ for all $l \in \{1, \dots, n\}$, notice that the linear map $T_1 = \sum_{i=1}^m \sum_{j=1}^n a_{ij} D_{ij}$ satisfies the following:

$$\begin{aligned} \forall l \in \{1, \dots, n\}, \quad T_1(v_l) &= \sum_{i=1}^m \sum_{j=1}^n a_{ij} D_{ij}(v_l) = \sum_{i=1}^m \sum_{j=1}^n a_{ij} \delta_{jl} w_i \\ &= \sum_{i=1}^m a_{il} w_i = T(v_l) \end{aligned} \quad (4.4)$$

So, since T, T_1 acts on the given basis $v_1, \dots, v_n \in V$ identically, $T = T_1$. Hence, $\text{Hom}_k(V, W) = \text{span}\{D_{ij}\}_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}}$.

Now, let $\varphi_1, \dots, \varphi_n \in V^\vee$ denotes the corresponding dual basis of $v_1, \dots, v_n \in V$ (i.e. $\varphi_j(v_l) = \delta_{jl}$), then notice that $\bar{B}(\varphi_j, w_i) = T_{\varphi_j, w_i}$ satisfies the following, for any $l \in \{1, \dots, n\}$:

$$T_{\varphi_j, w_i}(v_l) = \varphi_j(v_l) w_i = \delta_{jl} w_i = D_{ij}(v_l) \quad (4.5)$$

Hence, T_{φ_j, w_i} and D_{ij} acts on the given basis of V identically, showing $\bar{B}(\varphi_j, w_i) = T_{\varphi_j, w_i} = D_{ij}$. So, $\bar{B}$ has image containing $\text{span}\{D_{ij}\}_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}} = \text{Hom}_k(V, W)$, showing $\bar{B}$ is surjective, hence an isomorphism.

2. $\bar{B}$ is a morphism of G -representation: Let ρ_V, ρ_W denotes the representation of G with V, W respectively, and, let $\rho_\otimes$ denotes the representation of G with $V^\vee \otimes_k W$, while ρ_H denotes the representation of G with $\text{Hom}_k(V, W)$. Then, for all $g \in G$, for all $\varphi \in V^\vee$, $w \in W$, and $T \in \text{Hom}_k(V, W)$ we have the following action:

$$\rho_{\otimes}(g) \cdot (\varphi \otimes w) = (\varphi \circ \rho_V^{-1}(g)) \otimes (\rho_W(g) \cdot w), \quad \rho_{H(g)} \cdot T = \rho_W(g) \circ T \circ \rho_V^{-1}(g) \quad (4.6)$$

Then, notice that under $\overline{B}$, one has the following:

$$\begin{aligned} \overline{B}(\rho_{\otimes}(g) \cdot (\varphi \otimes w)) &= \overline{B}((\varphi \circ \rho_V(g)^{-1}) \otimes (\rho_W(g) \cdot w)) = T \\ \implies \forall v \in V, \quad T(v) &= (\varphi \circ \rho_V(g)^{-1} \cdot v) \rho_W(g)(w) \end{aligned} \quad (4.7)$$

$$\begin{aligned} \overline{B}(\varphi \otimes w) = T_{\varphi, w} \implies \forall v \in V, \quad (\rho_H(g) \cdot T_{\varphi, w})(v) &= \rho_W(g) \circ T_{\varphi, w} \circ \rho_V(g)^{-1}(v) \\ &= \rho_W(g)(\varphi(\rho_V(g)^{-1} \cdot v)w) \\ &= \varphi(\rho_V(g)^{-1} \cdot v) \rho_W(g)(w) \\ &= T(v) \end{aligned} \quad (4.8)$$

So, we can conclude that $\rho_H(g) \cdot \overline{B}(\varphi \otimes w) = \overline{B}(\rho_{\otimes}(g) \cdot (\varphi \otimes w))$, showing $\overline{B}$ is a homomorphism of G -representations.

So, this concludes that $V^{\vee} \otimes_k W \cong \text{Hom}_k(V, W)$ as G -representations.

5 D

Problem 5

Show that if two representations V and W have characters χ_1, χ_2 respectively, then the character of the tensor representation of V and W is given by $\chi_1 \chi_2$.

Solution: Here, fix a basis $v_1, \dots, v_n \in V$ and $w_1, \dots, w_m \in W$ (assume both are finite dimensional). Then, $V \otimes_k W$ have a basis $\{v_j \otimes w_l\}_{\substack{1 \leq j \leq n \\ 1 \leq l \leq m}}$.

Which, Let ρ_V, ρ_W denote the representation of G over V, W respectively. Let $\mathcal{M}(\rho_V(g)) = (a_{ij})_{1 \leq i, j \leq n}$, and $\mathcal{M}(\rho_W(g)) = (b_{kl})_{1 \leq k, l \leq m}$ under the given bases of V, W respectively. Then, one has $\rho_V(g)(v_j) = \sum_{i=1}^n a_{ij} v_i$, and $\rho_W(g)(w_l) = \sum_{k=1}^m b_{kl} w_k$.

Now, let ρ denotes the tensor representation of G over $V \otimes_k W$, then notice that its action on each basis of the tensor product is as follow:

$$\begin{aligned} \rho(g)(v_j \otimes w_l) &= (\rho_V(g)(v_j)) \otimes (\rho_W(g)(w_l)) = \left(\sum_{i=1}^n a_{ij} v_i \right) \otimes \left(\sum_{k=1}^m b_{kl} w_k \right) \\ &= \sum_{i=1}^n \sum_{k=1}^m a_{ij} b_{kl} (v_i \otimes w_k) \end{aligned} \quad (5.1)$$

So, the component of $v_j \otimes w_l$ after the action of g , is given by the component $i = j$ and $k = l$, which has coefficient $a_{jj} b_{ll}$.

Finally, if express $\rho(g)$ in the basis $\{v_j \otimes w_l\}_{\substack{1 \leq j \leq n \\ 1 \leq l \leq m}}$, then the diagonal entries are given by all the $v_j \otimes w_l$ -component of $\rho(g)(v_j \otimes w_l)$, which are all $a_{jj} b_{ll}$, where $1 \leq j \leq n$ and $1 \leq l \leq m$. So, we get the following:

$$\text{tr}(\rho(g)) = \sum_{j=1}^n \sum_{l=1}^m a_{jj} b_{ll} = \left(\sum_{j=1}^n a_{jj} \right) \left(\sum_{l=1}^m b_{ll} \right) = \text{tr}(\rho_V(g)) \cdot \text{tr}(\rho_W(g)) = \chi_1(g) \cdot \chi_2(g) \quad (5.2)$$

So, this indicates that the character of the tensor representation $V \otimes_k W$ is given by $\chi_1 \chi_2$.

6 D

Problem 6

Let V be a representation of G . Let $W \subset V \otimes V$ be the subspace of the tensor product $V \otimes V$ generated by vectors of the form $x \otimes y + y \otimes x$.

1. Show that W is a subrepresentation.
2. Suppose that V has dimension d , what is the dimension of W ?
3. Let χ denote the character of V , what is the character of W ?

Solution:

1. Let ρ_V denotes the representation of G over V , and ρ denotes the tensor representation of G over $V \otimes V$. Then, for all $x, y \in V$, and $g \in G$, the action is given as follow:

$$\rho(g)(x \otimes y) = (\rho_V(g)(x)) \otimes (\rho_V(g)(y)) \quad (6.1)$$

Hence, for the generators $x \otimes y + y \otimes x \in W$, the action is as follow:

$$\begin{aligned} \rho(g)(x \otimes y + y \otimes x) &= \rho(g)(x \otimes y) + \rho(g)(y \otimes x) \\ &= (\rho_V(g)(x)) \otimes (\rho_V(g)(y)) + (\rho_V(g)(y)) \otimes (\rho_V(g)(x)) \in W \end{aligned} \quad (6.2)$$

This shows that W is invariant under $\rho(g)$ (since all generators are still contained in W after $\rho(g)$ acts on it), hence W forms a subrepresentation of $V \otimes V$.

2. (Here, we'll assume $\text{char}(k) \neq 2$). Let $v_1, \dots, v_d \in V$ be a basis of V . We'll claim that $\dim W = \frac{d(d+1)}{2}$, by proving that the list $\{v_i \otimes v_j + v_j \otimes v_i\}_{1 \leq i \leq j \leq d} \subset W$ form a basis of W (Note: for all $j \in \{1, \dots, d\}$, there are j indices satisfy $i \leq j$, which the length of the list is $\sum_{j=1}^d j = \frac{d(d+1)}{2}$, which is the length of the list).

First, for all $x, y \in V$, it can be uniquely written as $x = \sum_{i=1}^n a_i v_i$ and $y = \sum_{j=1}^n b_j v_j$, hence we have the following:

$$\begin{aligned} x \otimes y + y \otimes x &= \left(\sum_{i=1}^n a_i v_i \right) \otimes \left(\sum_{j=1}^n b_j v_j \right) = \left(\sum_{j=1}^n b_j v_j \right) \otimes \left(\sum_{i=1}^n a_i v_i \right) \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i b_j (v_i \otimes v_j) + \sum_{i=1}^n \sum_{j=1}^n a_i b_j (v_j \otimes v_i) \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i b_j (v_i \otimes v_j + v_j \otimes v_i) \end{aligned} \quad (6.3)$$

If $i > j$, one can use $v_j \otimes v_i + v_i \otimes v_j$, then with $i' = j$ and $j' = i$, now the element becomes $v_{i'} \otimes v_{j'} + v_{j'} \otimes v_{i'}$, with $i' < j'$. Hence, this shows that $x \otimes y + y \otimes x \in \text{span}\{v_i \otimes v_j + v_j \otimes v_i\}_{i \leq j}$, or $W = \text{span}\{v_i \otimes v_j + v_j \otimes v_i\}_{i \leq j}$.

Then, to show linear independence, suppose $\sum_{i \leq j} a_{ij} (v_i \otimes v_j + v_j \otimes v_i) = 0$, notice that if $i \leq j$, the coefficient of $v_i \otimes v_j$ is a_{ij} , while if $i > j$, the coefficient of $v_i \otimes v_j$ is a_{ji} . Then, based on linear independence of the list $\{v_i \otimes v_j\}_{1 \leq i, j \leq n} \subset V \otimes V$, it enforces each $a_{ij}, a_{ji} = 0$. This shows the linear independence of the list.

Hence, we can conclude that $\dim W = \frac{d(d+1)}{2}$, with a convenient basis $\{v_i \otimes v_j + v_j \otimes v_i\}_{i \leq j}$.

3. Let χ denotes the character of V , then the character of $V \otimes V$ is χ^2 . Here, WLOG one can assume that $k = \bar{k}$ (the reason is if we're simply computing the trace, one can embed $\text{GL}_d(k) \hookrightarrow \text{GL}_d(\bar{k})$, which is possible to find an upper-triangular form in $\text{GL}_d(\bar{k})$ and make our life easier for computation, but still leave the trace invariant after the embedding).

For each $g \in G$, let $(a_{ij})_{1 \leq i, j \leq d}$ denotes the matrix representation of $\rho_V(g)$ under basis $\{v_i\}_{1 \leq i \leq n}$ (in particular, choose the basis so $\rho_V(g)$ is upper-triangular), then given basis $\{v_j \otimes v_l + v_l \otimes v_j\}_{j \leq l} \subset W$, one has $\rho(g)$ action as follow:

$$\begin{aligned} \rho(g)(v_j \otimes v_l) &= (\rho_V(g) \cdot v_j) \otimes (\rho_V(g) \cdot v_l) = \left(\sum_{i=1}^n a_{ij} v_i \right) \otimes \left(\sum_{k=1}^n a_{kl} v_k \right) \\ &= \sum_{i=1}^n \sum_{k=1}^n a_{ij} a_{kl} (v_i \otimes v_k) = \sum_{i=1}^j \sum_{k=1}^l a_{ij} a_{kl} (v_i \otimes v_k) \end{aligned} \quad (6.4)$$

(With the restriction of upper-triangular matrix). So, we get:

$$\rho(g)(v_j \otimes v_l + v_l \otimes v_j) = \sum_{i=1}^j \sum_{k=1}^l a_{ij} a_{kl} (v_i \otimes v_k) + \sum_{i'=1}^l \sum_{k'=1}^j a_{i'l} a_{k'j} (v_{i'} \otimes v_{k'}) \quad (6.5)$$

This indicates that the $(v_j \otimes v_l + v_l \otimes v_j)$ component of $\rho(g)(v_j \otimes v_l + v_l \otimes v_j)$ is given by $i = j, k = 1$ in the first sum, and $i' = l, l' = j$ in the second sum, which provides the term $a_{jj} a_{ll} (v_j \otimes v_l) + a_{ll} a_{jj} (v_l \otimes v_j) = a_{jj} a_{ll} (v_j \otimes v_l + v_l \otimes v_j)$ (Note: since the restriction is $j \leq l$, so there's no other components missing).

Hence, under this basis of W , we have the following:

$$\begin{aligned} \text{tr} \Big|_W (\rho(g)) &= \sum_{j \leq l} a_{jj} a_{ll} = \frac{1}{2} \left(2 \cdot \sum_{j \leq l} a_{jj} a_{ll} \right) = \frac{1}{2} \left(\sum_{j,l} a_{jj} a_{ll} + \sum_{j=l} a_{jj} a_{ll} \right) \\ &= \frac{1}{2} \left(\sum_{j=1}^d \sum_{l=1}^d a_{jj} a_{ll} + \sum_{i=1}^d a_{ii}^2 \right) = \frac{1}{2} \left(\left(\sum_{j=1}^d a_{jj} \right) \left(\sum_{l=1}^d a_{ll} \right) + \sum_{i=1}^d a_{ii}^2 \right) \end{aligned} \quad (6.6)$$

Now, notice that over V , one has $\chi(g) = \text{tr}(\rho_V(g)) = \sum_{j=1}^d a_{jj}$, so $\left(\sum_{j=1}^d a_{jj} \right) \left(\sum_{l=1}^d a_{ll} \right) = \chi(g)^2$; also, since $\rho_V(g)$ is constructed to be upper-triangular, then diagonals of $\rho_V(g^2) = \rho_V(g)^2$ are precisely given by a_{ii}^2 . So, we have $\chi(g^2) = \text{tr}(\rho_V(g^2)) = \sum_{i=1}^d a_{ii}^2$. So, we get the following:

$$\text{tr} \Big|_W (\rho(g)) = \frac{\chi(g)^2 + \chi(g^2)}{2} \quad (6.7)$$

7 D

Problem 7

Determine all (isomorphism classes of) irreducible representations of S_3 and D_8 .

Solution: First, we'll recall that for a finite group G with its representations over $\mathbb{C}$ -vector spaces, if $(V_1, \rho_1), \dots, (V_k, \rho_k)$ are mutually non-isomorphic irreducible representations of G , then with $d_i := \dim V_i$ for each index i , one has $|G| = \sum_{i=1}^k d_i^2$; also, $k =$ number of conjugacy classes in G .

1. For S_3 , since $|S_3| = 6$, and in **Problem 2** we've computed all conjugacy classes: $C_{1,1,1} = \{e\}$, $C_{1,2} = \{(12), (13), (23)\}$, and $C_3 = \{(123), (132)\}$. Then, the only way of partition 6 into three perfect squares, is $6 = 1 + 1 + 2^2$, this indicates that S_3 has two 1-dimensional, and one 2-dimensional irreducible representations.

For the 1-dimension cases, given $\mathbb{C}^\times = \text{GL}_1(\mathbb{C})$, there are two immediate group homomorphisms $S_3 \rightarrow \mathbb{C}^\times$, given as follow:

- The trivial map $\mathbb{1} : S_3 \rightarrow \mathbb{C}^\times$ by $\mathbb{1}(\sigma) = 1$ forms the trivial representation.
- The sign map $\text{sign} : S_3 \rightarrow \{\pm 1\} \subset \mathbb{C}^\times$ by $\sigma \mapsto \text{sign}(\sigma)$ forms the sign representation (which is distinct from trivial representation, since $\text{sign}((12)) = -1$, while $\mathbb{1}((12)) = 1$).

For the 2-dimensional case, we'll find an irreducible subrepresentation in $\mathbb{C}^3$: Given standard basis e_1, e_2, e_3 (representing $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ respectively). Let S_3 define an action on $\mathbb{C}^3$ by $\sigma(xe_1 + ye_2 + ze_3) := xe_{\sigma(1)} + ye_{\sigma(2)} + ze_{\sigma(3)}$ (i.e. permute the ordered coordinates).

First, notice that the 2-dimensional subspace $V := \{(x, y, z) \in \mathbb{C}^3 \mid x + y + z = 0\}$ is a subrepresentation, since regardless of the permutation, the three entries are always given by x, y, z (may be in some other order), so before and after the permutation, the sum of the entries is always $x + y + z = 0$, showing that V is invariant under S_3 -action on $\mathbb{C}^3$, which V is a subrepresentation.

Now, we claim that V is irreducible: Suppose the contrary that V is reducible, meaning it has a proper nontrivial subrepresentation $W \subset V$, enforcing W to be 1-dimensional, or $W = \text{span}\{(x, y, z)\}$ for some $(x, y, z) \in V$. However, if it's a subrepresentation, it must be isomorphic to one of the irreducible 1-dimensional representations mentioned before, in either case one always have 3-cycle (123) acts as an identity on the vector space. So, we must have the following:

$$(x, y, z) = (123) \cdot (x, y, z) = (z, x, y) \implies x = y = z \quad (7.1)$$

However, this implies that $3x = x + y + z = 0$, or $0 = x = y = z$, showing $(x, y, z) = (0, 0, 0)$, which contradicts the assumption that W is a 1-dimensional subspace. So, V must be irreducible, hence the desired 2-dimensional irreducible representation of S_3 .

2. For D_8 , since $|D_8| = 8$, and in **Problem 2** we've also computed all conjugacy classes: $C_1 = \{e\}$, $C_2 = \{\sigma^2\}$, $C_3 = \{\sigma, \sigma^3\}$, $C_4 = \{\tau, \tau\sigma^2\}$, and $C_5 = \{\tau\sigma, \tau\sigma^3\}$. Then, the only way of partition 8 into five perfect squares, is $8 = 1 + 1 + 1 + 1 + 2^2$, this indicates that D_8 has four 1-dimensional, and one 2-dimensional irreducible representations.

For the 1-dimension cases, given any representation $\varphi : D_8 \rightarrow \mathbb{C}^\times = \text{GL}_1(\mathbb{C})$, for any $g \in D_8$, we have its character $\chi_\varphi(g) = \text{tr}(\varphi(g)) = \varphi(g)$ (since it's a 1×1 matrix), then $\chi_\varphi =$

φ as a set map. Then, with χ_φ being conjugation invariant, so is φ . Hence, based on the conjugacy classes, we have the following:

- Based on relation of C_3 , $\varphi(\sigma) = \varphi(\sigma^3) = \varphi(\sigma)^3$, so $\varphi(\sigma^2) = \varphi(\sigma)^2 = 1$, showing $\varphi(\sigma) = \pm 1$.
- For relation on τ , since $\varphi(\tau)^2 = \varphi(\tau^2) = \varphi(e) = 1$, then $\varphi(\tau) = \pm 1$ also.

Which, there are four possible combinations: either $\varphi(\sigma) = \varphi(\tau) = 1$, $\varphi(\sigma) = 1$ and $\varphi(\tau) = -1$, $\varphi(\sigma) = -1$ and $\varphi(\tau) = 1$, or $\varphi(\sigma) = \varphi(\tau) = -1$. Using $\varphi_1, \varphi_{\pm 1}, \varphi_{\mp 1}, \varphi_{-1}$ respectively to denote the corresponding maps, we have the following results:

$$\varphi_1(D_8) = \{1\} \quad (7.2)$$

$$\begin{aligned} \varphi_{\pm 1}(e) &= \varphi_{\pm 1}(\sigma^2) = \varphi_{\pm 1}(\sigma) = \varphi_{\pm 1}(\sigma^3) = 1 \\ \varphi_{\pm 1}(\tau) &= \varphi_{\pm 1}(\tau\sigma^2) = \varphi_{\pm 1}(\tau\sigma) = \varphi_{\pm 1}(\tau\sigma^3) = -1 \end{aligned} \quad (7.3)$$

$$\begin{aligned} \varphi_{\mp 1}(e) &= \varphi_{\mp 1}(\sigma^2) = \varphi_{\mp 1}(\tau) = \varphi_{\mp 1}(\tau\sigma^2) = 1 \\ \varphi_{\mp 1}(\sigma) &= \varphi_{\mp 1}(\sigma^3) = \varphi_{\mp 1}(\tau\sigma) = \varphi_{\mp 1}(\tau\sigma^3) = -1 \end{aligned} \quad (7.4)$$

$$\begin{aligned} \varphi_{-1}(e) &= \varphi_{-1}(\sigma^2) = \varphi_{-1}(\tau\sigma) = \varphi_{-1}(\tau\sigma^3) = 1 \\ \varphi_{-1}(\sigma) &= \varphi_{-1}(\sigma^3) = \varphi_{-1}(\tau) = \varphi_{-1}(\tau\sigma^2) = -1 \end{aligned} \quad (7.5)$$

These are all distinct maps, and one can check that all of them are group homomorphisms $D_8 \rightarrow \{\pm 1\} \subset \mathbb{C}^\times$, which forms all the 1-dimensional group representations of D_8 .

For the 2-dimension case, we'll again find a subrepresentation in $\mathbb{C}^4$ (with standard basis e_1, e_2, e_3, e_4): If consider the dihedral group D_8 as the symmetry group on a square in $\mathbb{C}$ (with four vertices labeled as 1, 2, 3, 4 in counterclockwise order, and with the vertex 1 lying on positive real axis; for convenience 1, 2, 3, 4 can also be represented with 1, i , -1 , $-i$ using coordinates on $\mathbb{C}$), then let σ denotes the counterclockwise rotation, τ denotes the reflection by imaginary axis, we have $\sigma = (1234)$, and $\tau = (13)$ as permutations of the vertices. Hence, this creates an embedding $D_8 \hookrightarrow S_4$.

Now, consider the natural action of S_4 on $\mathbb{C}^4$ (which induces a subgroup representation of D_8 over $\mathbb{C}^4$), by $\sigma(xe_1 + ye_2 + ze_3 + we_4) := xe_{\sigma(1)} + ye_{\sigma(2)} + ze_{\sigma(3)} + we_{\sigma(4)}$, notice that the 3-dimensional subspace $V = \{(x, y, z, w) \in \mathbb{C}^4 \mid x + y + z + w = 0\}$ forms a subrepresentation of S_4 (since after permuting the coordinates, the entries are still given by x, y, z, w , so the sum $x + y + z + w = 0$ is invariant). Hence, with $D_8 \hookrightarrow S_4$, V is also a subrepresentation of D_8 .

Second, consider the vector $(1, -1, 1, -1) \in V$ (since $1 + (-1) + 1 + (-1) = 0$). Notice that under rotation $\sigma = (1234)$ and reflection $\tau = (13)$, they have the following:

$$\sigma(1, -1, 1, -1) = (-1, 1, -1, 1) = -(1, -1, 1, -1), \quad \tau(1, -1, 1, -1) = (1, -1, 1, -1) \quad (7.6)$$

Hence, the 1-dimensional subspace $W = \text{span}\{(1, -1, 1, -1)\} \subset V$ is invariant under the action of D_8 's generators, hence invariant under D_8 . This shows that V has a 1-dimensional subrepresentation.

Then, consider the orthogonal complement of W in V , denoted as U , then $U = \{(x, y, z, w) \in \mathbb{C}^4 \mid x + y + z + w = 0, \quad x - y + z - w = 0\}$ (the second equation is the inner product with $(1, -1, 1, -1)$). For any $(x, y, z, w) \in U$, if consider the action of σ and τ on it, we have the following:

$$\sigma(x, y, z, w) = (w, x, y, z), \quad w - x + y - z = -(x - y + z - w) = 0 \quad (7.7)$$

$$\tau(x, y, z, w) = (z, y, x, w), \quad z - y + x - w = x - y + z - w = 0 \quad (7.8)$$

Hence, $\sigma(x, y, z, w), \tau(x, y, z, w) \in U$, showing that U is invariant under D_8 's action (since it's invariant under the generators' actions). Hence, U is a 2-dimensional subrepresentation of D_8 .

Finally, we claim that U is irreducible: Again, suppose the contrary that U is reducible, then there exists a proper nontrivial subrepresentation $Z \subset U$, which enforces Z to be 1-dimensional. Now, say $Z = \text{span}\{(x, y, z, w)\}$ (which by definition of U , it satisfies $w = -x - y - z$, and $0 = x - y + z - w = x - y + z + (x + y + z) = 2x + 2z$, enforcing $x = -z$, which further implies $w = -x - y - z = -y$), then under the action of σ and τ , by the invariance of Z under their actions, there exists $\lambda, \eta \in \mathbb{C}$, such that the following is true:

$$\lambda(-z, y, z, -y) = \sigma(-z, y, z, -y) = (-y, -z, y, z) \implies z = -\lambda y, y = \lambda z \quad (7.9)$$

$$\eta(-z, y, z, -y) = \tau(-z, y, z, -y) = (z, y, -z, -y) \implies y = \eta y, z = -\eta z \quad (7.10)$$

The equation $y = \eta y$ implies that either $y = 0$ (which implies $z = -\lambda y = 0$, so $(-z, y, z, -y) = 0$, which contradicts the fact that $Z = \text{span}\{(-z, y, z, -y)\}$ is 1-dimensional), or $\eta = 1$. But, $z = -\eta z = -z$ implies that $z = 0$, which further implies $y = \lambda z = 0$, which is again a contradiction.

So, U must be irreducible, hence the desired 2-dimensional irreducible representation of D_8 .

8 D

Problem 8

Compute the character tables for S_3 and D_8 .

Solution: In both cases, in **Problem 7** we've proven that for 1-dimensional representations, the representation $\rho : G \rightarrow \mathbb{C}^\times$ is the same as the character $\chi_\rho : G \rightarrow \mathbb{C}^\times$. Hence, we can directly use the results from the previous problem for all 1-dimensional case. For practice purpose, we'll practice using orthonormality to solve for the remaining entries.

1. Given S_3 , let χ_1, χ_2 denotes the 1-dimensional irreducible characters, and χ_3 denotes the 2-dimensional irreducible characters. For any character χ , since $\chi(e) - \text{tr}(\text{id}_V) = \dim(V)$, then for the conjugacy class $C_{1,1,1} = \{e\}$, it must be evaluated to be the dimension of the corresponding representation. Hence, we have the following table:

Character \ Class	$C_{1,1,1}$	$C_{1,2}$	C_3
χ_1	1	1	1
χ_2	1	-1	1
χ_3	2	x	y

Then, using orthonormality, we have the following (recall that $|C_{1,1,1}| = 1$, $|C_{1,2}| = 3$, and $|C_3| = 2$):

$$\langle \chi_3, \chi_1 \rangle = \frac{1}{6} \cdot (2 \cdot 1 + x \cdot 3 + y \cdot 2) = 0 \implies 2 + 3x + 2y = 0 \quad (8.1)$$

$$\langle \chi_3, \chi_2 \rangle = \frac{1}{6} \cdot (2 \cdot 1 + x \cdot (-3) + y \cdot 2) = 0 \implies 2 - 3x + 2y = 0 \quad (8.2)$$

Adding the two equations, we get that $4 + 4y = 0$, or $y = -1$, which results in $2 + 3x + 2(-1) = 0$, so $3x = 0$, or $x = 0$. Hence, the full table is as follow:

Character \ Class	$C_{1,1,1}$	$C_{1,2}$	C_3
χ_1	1	1	1
χ_2	1	-1	1
χ_3	2	0	-1

2. Given D_8 , let $\chi_1, \chi_2, \chi_3, \chi_4$ denote the 1-dimensional irreducible characters, and χ_5 denotes the 2-dimensional irreducible characters. Together with the dimension claim, and the 1-dimensional character being the same as the group representation, we get the following table (where the characters correspond to the representations $\varphi_1, \varphi_{\pm 1}, \varphi_{\mp 1}, \varphi_{-1}$ respectively):

Character \ Class	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	1	-1	1	-1
χ_4	1	1	-1	-1	1
χ_5	2	x	y	z	w

Then, based on the orthonormality, we get the following four equations (Note: $|C_1| = |C_2| = 1$, and $|C_3| = |C_4| = |C_5| = 2$):

$$\langle \chi_5, \chi_1 \rangle = 2 \cdot 1 + x \cdot 1 + y \cdot 2 + z \cdot 2 + w \cdot 2 = 0 \implies 2 + x + 2y + 2z + 2w = 0 \quad (8.3)$$

$$\langle \chi_5, \chi_2 \rangle = 2 \cdot 1 + x \cdot 1 + y \cdot 2 - z \cdot 2 - w \cdot 2 = 0 \implies 2 + x + 2y - 2z - 2w = 0 \quad (8.4)$$

$$\langle \chi_5, \chi_3 \rangle = 2 \cdot 1 + x \cdot 1 - y \cdot 2 + z \cdot 2 - w \cdot 2 = 0 \implies 2 + x - 2y + 2z - 2w = 0 \quad (8.5)$$

$$\langle \chi_5, \chi_4 \rangle = 2 \cdot 1 + x \cdot 1 - y \cdot 2 - z \cdot 2 + w \cdot 2 = 0 \implies 2 + x - 2y - 2z + 2w = 0 \quad (8.6)$$

Add the first two (and respectively, the last two) equations together, we get:

$$4 + 2x + 4y = 0, \quad 4 + 2x - 4y = 0 \implies 8 + 4x = 0 \implies x = -2 \quad (8.7)$$

Plug into the equation, we get:

$$4 + 2(-2) + 4y = 0 \implies 4y = 0 \implies y = 0 \quad (8.8)$$

Now, plug $x = -2, y = 0$ into other equations, we get:

$$\begin{aligned} 2 + (-2) + 2z + 2w &= 0 \implies z + w = 0 \\ 2 + (-2) + 2z - 2w &= 0 \implies z - w = 0 \end{aligned} \quad (8.9)$$

Hence, add these two equations up, it derives $2z = 0$, or $z = 0$; then, this results in $w = 0$ also. So, the full table is as follow:

Character\Class	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	1	-1	1	-1
χ_4	1	1	-1	-1	1
χ_5	2	-2	0	0	0